Thiranex Internship — Task 4

End-to-End Retail Analytics | Project Report

Prepared by: Pratyush **Dataset:** Synthetic educational sample

1. Aim and methodology

An end-to-end analysis of fictional e-commerce transactions using customer, product and sales tables. Steps include checking data quality, validated relational joins, calculating net revenue after discount, grouping transactions by month, category, product and region, and visualizing performance.

2. Summary of findings

Metric	Computed value
Transactions	1,800
Unique customers	250
Total revenue	INR 3,606,930.50
Average order value	INR 2,003.85
Highest revenue month	2025-11
Top category	Fashion
Top product	Jacket
Largest revenue region	West

3. Interpretation and limitations

The data show an illustrative seasonal increase in late-year sales and differences across product categories and regions. Because all records were generated synthetically, these patterns cannot support real business recommendations or causal claims. No financial forecasting model is claimed.

Figure 1: Monthly Revenue

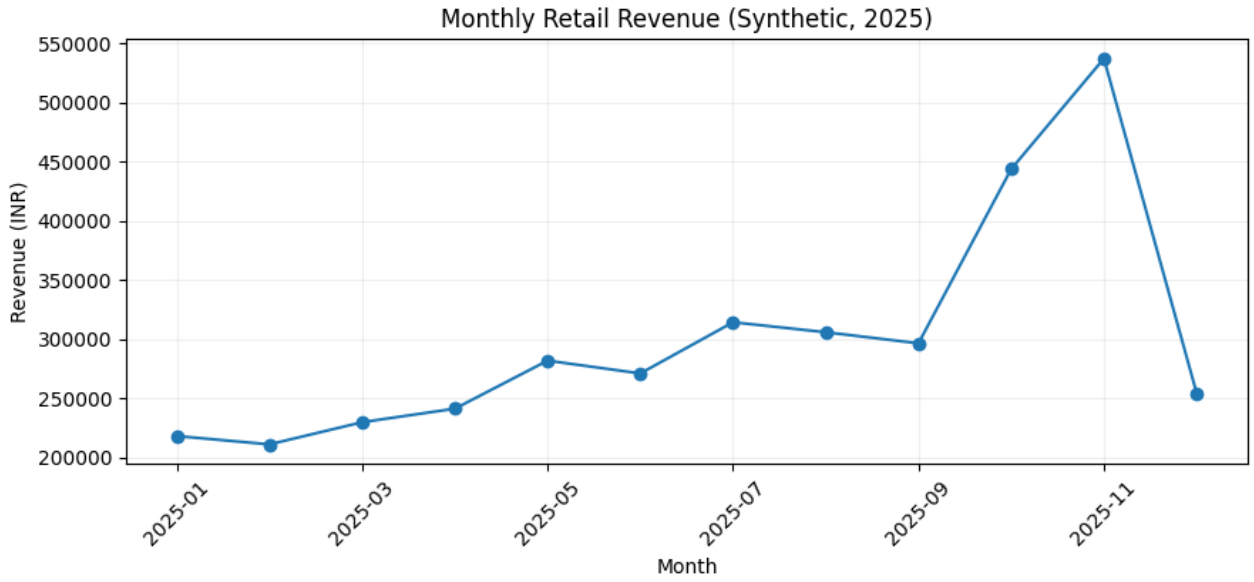


Figure 2: Category Revenue

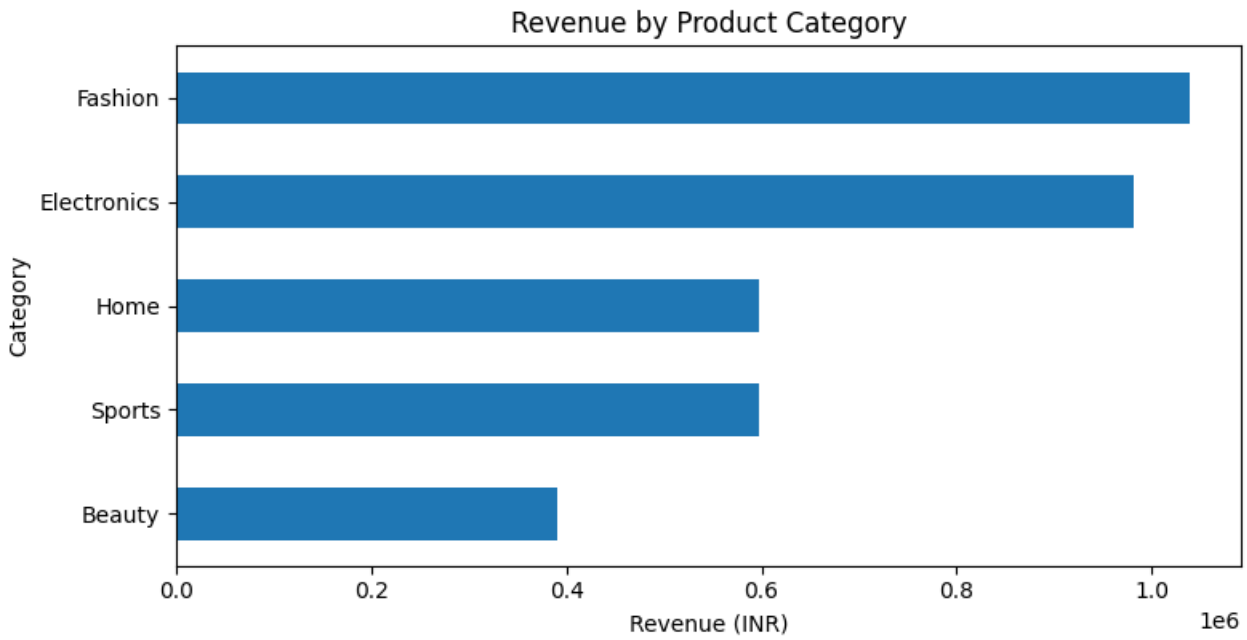


Figure 3: Top Products

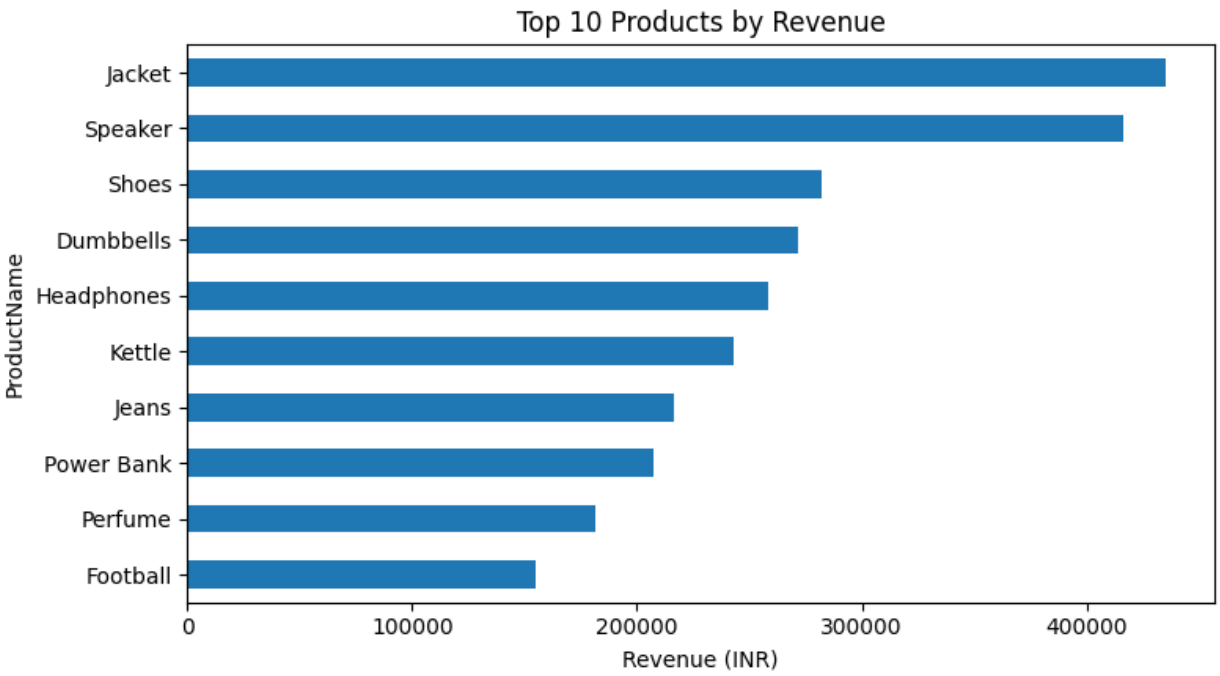


Figure 4: Region Revenue

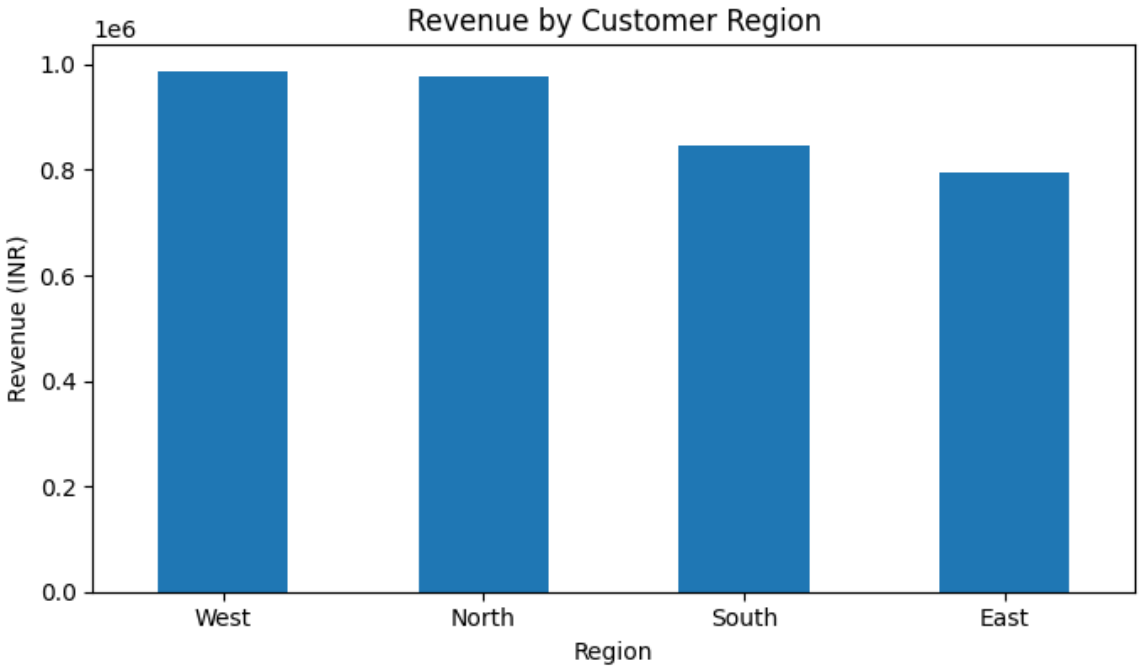


Figure 5: Customer Spending

